portkit for the bitbox micro: manual

This manual covers what portkit adds to the 1010music **bitbox micro** firmware 2.3.4 (build **2.3.4-pkZD**). Everything else works as in the stock firmware; see 1010music's own manual for that. Also as a PDF: portkit-micro-manual.pdf.

The screenshots come from the simulation of the patched firmware, so they show exactly this build. On the unit the screen looks the same; the pad names here are just the file names of the demo card.

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1. Install and check

1. Get the stock firmware **2.3.4** from 1010music and take `MICRO.BIN` out of the zip. Keep a copy.
2. Open the patcher page, drop the stock `MICRO.BIN` on it and download the patched `MICRO.BIN`. It runs in your browser; nothing is uploaded.
3. Copy the patched `MICRO.BIN` to the root of the microSD card.
4. Hold the **white right arrow** key while switching the unit on.



Splash screen: bitbox micro, portkit mod, 2.3.4-pkZD

The splash screen shows **portkit mod** and the version **2.3.4-pkZD**. The version is also at the top of the Tools settings page.

2. Getting around

portkit uses the unit's normal controls:

Control	What it does
HOME	cycles the main screens: Pads → Mixer → Effects (FX)
TOOLS	cycles the Tools screens: Settings → Input → Output
Left / right arrow	pages within a screen (pad pages, master pages, pads 1–8 / 9–16)
Touch	select a pad, a cell or a list row; swipe a list to scroll it
Encoder	change the selected value

A pad's settings: select the pad, then press the **right arrow** (waveform), and again (parameter list). The **page menu** icon (the dots, second from the left in the pad header) opens the list of parameter pages:



Pad page menu: Main, LFO, Env, Config, Pos

Page	portkit additions on it
Main	Filter 2, Res 2, Drive (all pad modes); the 24 dB filter
Config	Fidelity (renamed from Interp, with the vintage sampler modes)
Pos (Clip pads)	Shift ms, Play Len, Stretch %, Str Mode, Loop Fade; Sync 1/32

3. Pads 9–16

The pad grid has a second page: pads 9–16.



Pads screen, pads 1-8



Pads screen, pads 9-16

- **Pads screen:** the **left arrow** switches between *Pads 1-8* and *Pads 9-16* (shown in the title).
- **Effects screen:** the left arrow switches its pads the same way (*FX 1-8* / *FX 9-16*).
- **Mixer:** tap the field at the top left.
- **Pad picker** (the grid icon at the left of a pad's header): the left arrow shows the other page.



Pad picker, page 1



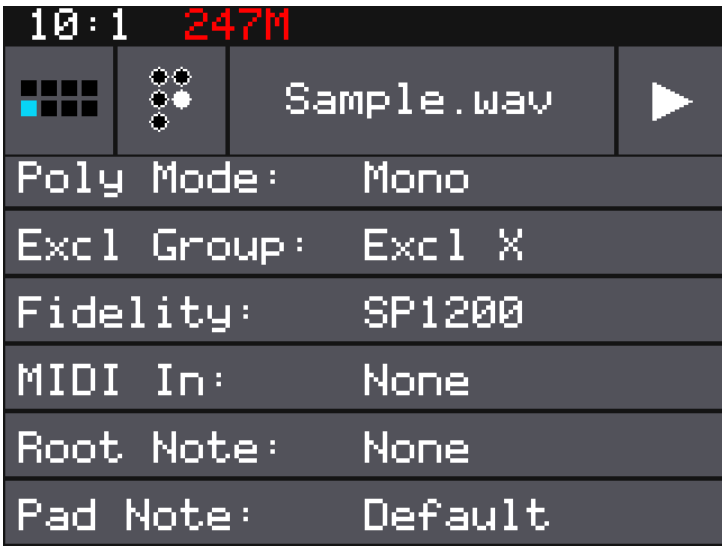
Pad picker, page 2

Pads on page 2 work like pads 1–8 and are saved with the preset.

4. Sound: Fidelity, Drive, Filter 2

Fidelity

On every pad mode (Sample, Clip, Slicer, Granular): page menu → **Config** → **Fidelity**.

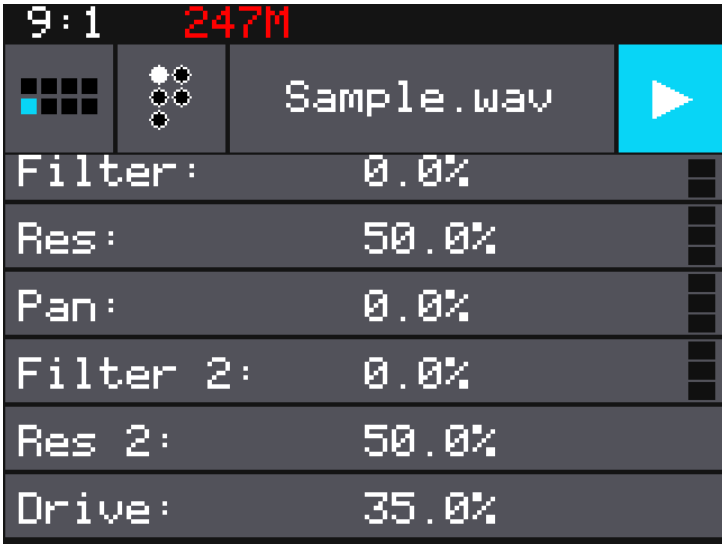


Config page: Fidelity SP1200

Setting	Sound
Normal, HighQ	as stock (stock interpolation)
SP1200, SP-12, S950, MPC60, SP Raw	machine modes voiced after those vintage samplers
16k, 12k, 10k, 8k, 6k, 4k, 2k	plain sample-rate reduction to that rate, 12 bit, no anti-aliasing filter

The reduction happens before the pad filter, so the filter can tame the grit.

24 dB filter, Drive, Filter 2



Main page: Filter, Res, Pan, Filter 2, Res 2, Drive

- **Filter / Res** (stock controls) now drive a 24 dB/octave state-variable filter instead of the stock 12 dB one: left of centre low-pass, right of centre high-pass, centre off. The resonance stays clean.
- **Drive**: soft saturation (tanh) before the filters, 0..100 %.
- **Filter 2 / Res 2**: a second 24 dB filter in series, same behaviour as Filter / Res. Use both for a steeper slope or a band-pass (one low-pass, one high-pass).

Scroll the Main page list (swipe up) to reach Filter 2, Res 2 and Drive. Old presets open with Drive 0 and Filter 2 off.

5. Clips: Shift, Play Len, Stretch, Loop Fade, Sync 1/32

On a **Clip** pad: page menu → **Pos**, then scroll down.



Clip Pos page: Sync 1/16, Shift 30, Play Len Full, Stretch 25, Str Mode Smooth, Loop Fade 10

Control	Range	What it does
Sync	adds **1/32**	finer re-trigger grid for clips
Shift ms	−100..+100 ms	moves the audio against the loop: the loop keeps its length and grid position, its content rotates. For loops that start slightly early or late
Play Len	Full, 1/4..32 bars	plays only that length of the clip, in quarter-bar steps
Stretch %	−100..+100 %	plays the clip slower or faster than its natural tempo
Str Mode	Auto / Jump / Smooth	how Stretch works: *Jump* skips through the clip in segments (keeps transients, can stutter), *Smooth* time-stretches without changing the pitch (WSOLA), *Auto* = Smooth whenever Stretch is not 0, otherwise the stock playback
Loop Fade	0..50 ms	a short dip at the loop start, loop end and the Shift seam against clicks

Play Len and Stretch need **Sync**: with Sync set to *Slice* or *None* they have no effect and are hidden, as are they when Quant Size makes them ineffective.

6. Effects: reverb types and FX3

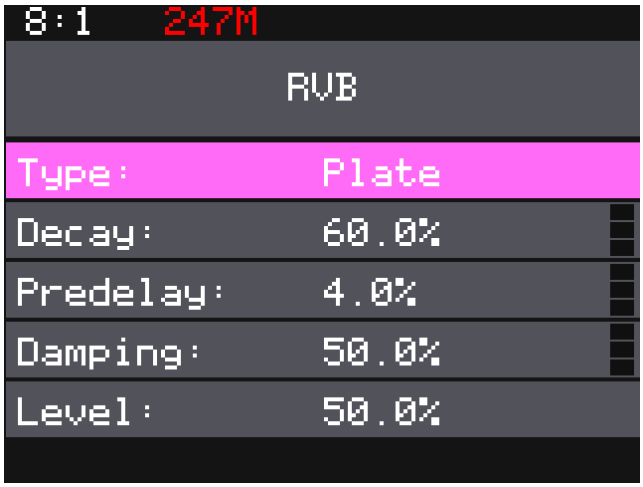
Press **HOME** until the Effects screen (*FX 1-8*) shows. The top row holds the effect cells, below are the pads.

Reverb types

Tap **RVB**, then the **right arrow**:



Effects screen, RVB selected



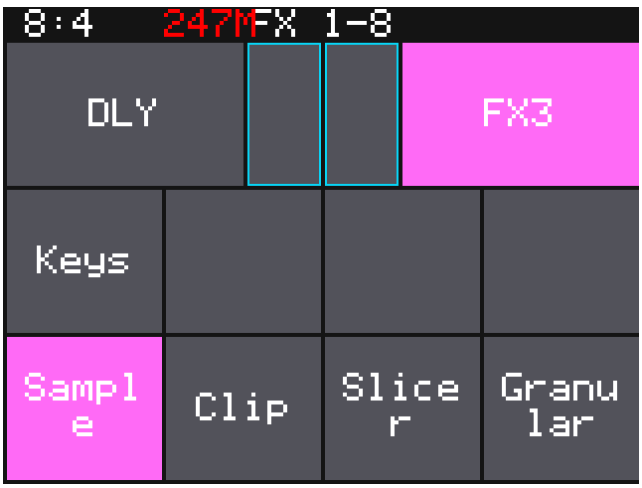
Reverb page: Type Plate, Decay, Predelay, Damping, Level

Type: Plate (stock), Room, Hall, Chamber, each with its own settings (scroll the list) and a **Level** control. The types are level-matched to the plate.

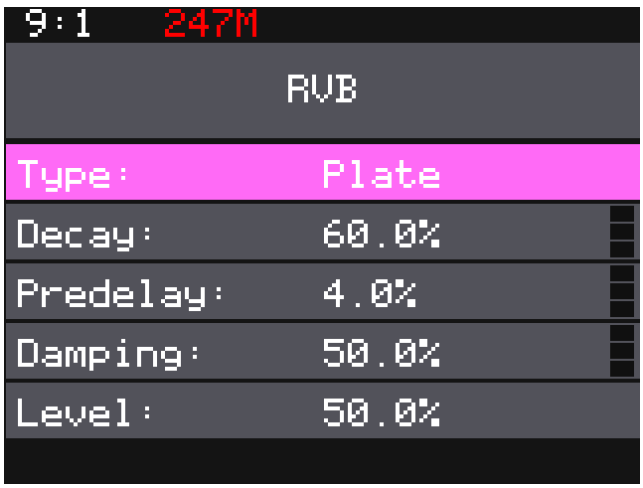
FX3: a third send

FX3 is a second, independent reverb on its own send, e.g. a short room on the drums and a long hall on the synths.

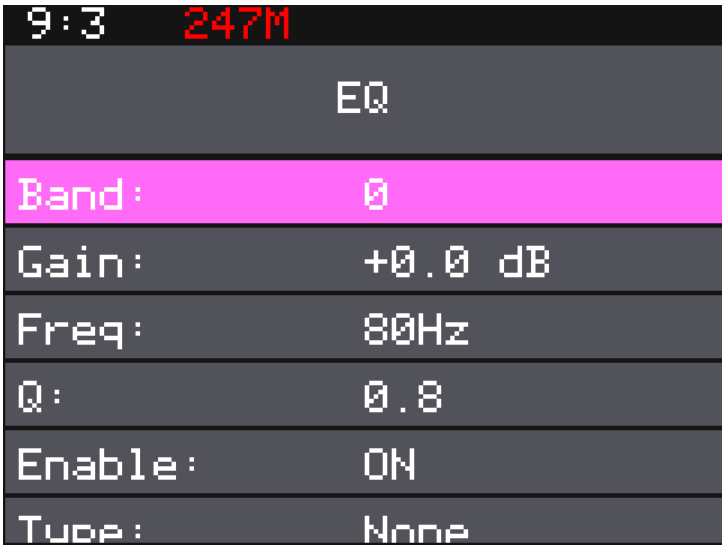
- On the Effects screen, **tap the selected RVB cell again**: it switches to **FX3** (and back).
- While FX3 is shown, the pads below show and set each pad's **FX3 send**, as they do for DLY and RVB.
- **Right arrow** opens the FX3 settings. The page is titled with the reverb type (FX3 is a reverb), it has the same types and settings as RVB.
- **Right arrow again** opens the **master EQ**.



Effects screen showing FX3



FX3 settings

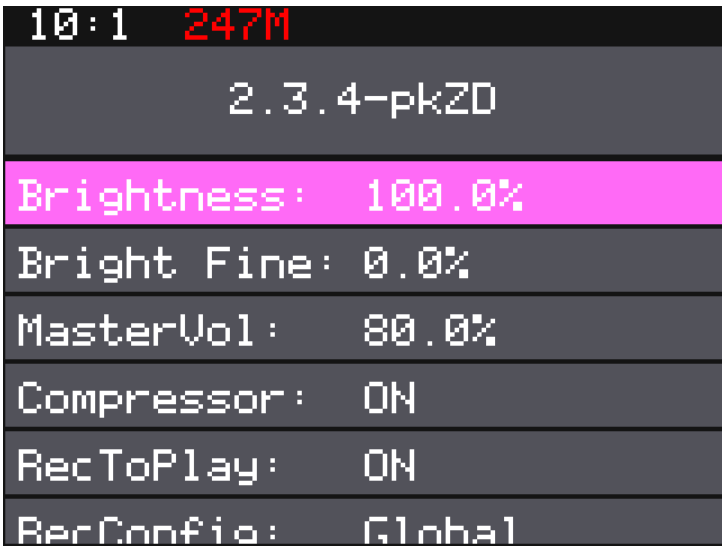


Master EQ page

7. Master section: compressor and distortion

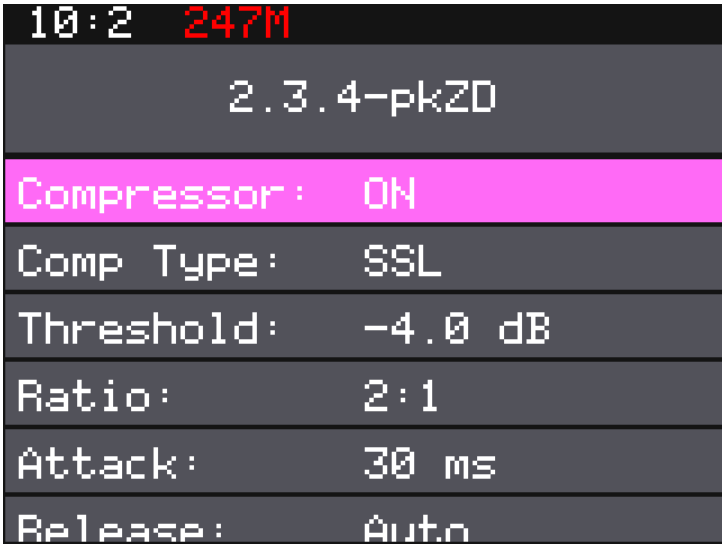
Press **TOOLS** for the Settings page (the version is at the top). From there:

- **Left arrow**: the **compressor page**
- **Right arrow** (from the compressor page, twice): the **distortion page**



Tools settings page with version 2.3.4-pkZD

Compressor

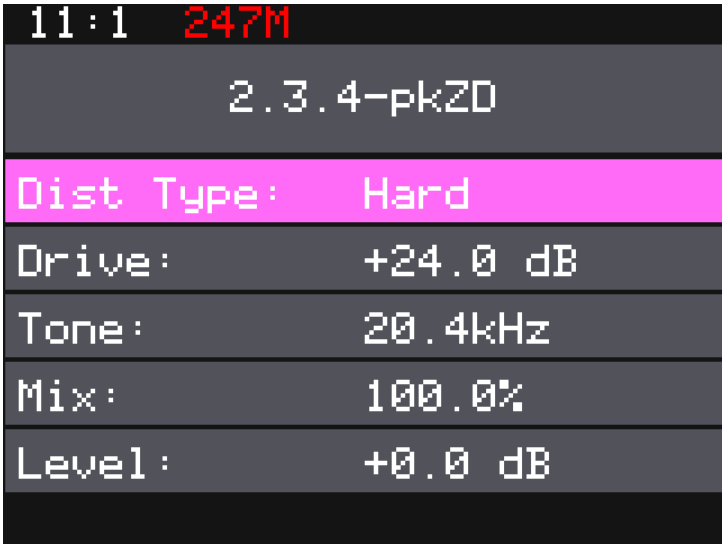


Compressor page: Compressor ON, Comp Type SSL, Threshold, Ratio, Attack, Release

Control	Stock type	SSL type
Compressor	on / off (the stock switch)	
Comp Type	Stock: the firmware's own compressor	SSL: a bus compressor modelled on the SSL console's
Threshold	dB	dB
Ratio	1:1 .. 20:1	2:1, 4:1, 10:1
Attack	1 .. 200 ms	0.1 / 0.3 / 1 / 3 / 10 / 30 ms
Release	10 .. 2000 ms	0.1 / 0.3 / 0.6 / 1.2 s or **Auto** (program dependent)
Gain	make-up gain, −12 .. +24 dB	same
Mix	dry / compressed blend (parallel compression)	same

The compressor works on the main output after the effects and the distortion.

Master distortion



Distortion page: Dist Type Hard, Drive +24 dB, Tone, Mix, Level

Control	What it does
Dist Type	Off, Soft (tanh saturation), Hard (clipping), Fold (triangle wave folding), Tube (asymmetric saturation)
Drive	input gain into the distortion, 0 .. +36 dB
Tone	low-pass after the distortion, fully open at the top (20.4 kHz)
Mix	dry / distorted blend
Level	output level, −24 .. +12 dB

It runs with 2x oversampling on the main output (Out 1/2), after the effect returns and the master EQ and before the compressor, whether the compressor is on or off.

8. Recording

Recording works as in the stock firmware, with two changes:

- A pad's **FX1 / FX2 / FX3 sends stay** as they were when you record into it (the stock firmware reset them).
- The **input monitor** goes through the sends of the selected pad, so you hear the input with the effects it will have.

9. Presets, going back, troubleshooting

- **Presets:** values are stored by name. Presets from the stock firmware load as before; the new controls start at their defaults (Fidelity Normal, Drive 0, Filter 2 off, Shift 0, Play Len Full, Stretch 0, FX3 send 0, distortion Off, compressor type Stock).
- **Going back to stock:** the same install steps with the stock `MICRO.BIN`. Presets saved with portkit contain extra values the stock firmware does not know; how stock treats them has not been tested, so keep a copy of presets you care about.
- **The patcher page says "not the stock MICRO.BIN":** the file is not the unmodified 2.3.4 (another version, or already patched). Take a fresh copy out of 1010music's 2.3.4 zip.
- **Play Len / Stretch are missing on the Pos page:** set Sync to a note value (not Slice or None).
- **Problems:** please open an issue on GitHub with the build (2.3.4-pkZD) and what you did.

Screenshots: `python3 -I tools/manual_shots.py` *(Renode simulation of the patched build). PDF:* `tools/manual_pdf.py`. *Unofficial community firmware, not affiliated with 1010music.*